



The Pleistocene and the Paleolithic

Archaeologists use a number of chronological terms to describe the periods in which human evolved, but this array of terminology can be confusing when you begin to learn about these periods. The most important idea to keep in mind is that archaeologists use both archaeological and geological terms to describe specific periods. As shown in the chart below, there is often a correlation between these two sets of terms. For example, the geological era of the **Pleistocene** overlaps with the archaeological period known as the **Paleolithic**, or “Old Stone Age.” Both are divided into three subperiods—termed Upper, Middle, and Lower—from most recent to oldest (although note that the periods of the Pleistocene are often termed Early, Middle, and Late instead). Confusingly, these do not exactly overlap. This is because the archaeological periods are based on technological innovations in human behavior, while the geological periods are based on climatic changes. Notably, both came to an end around 12,000 years ago, when we entered the current geological period, the Holocene, which is markedly warmer, wetter, and more climatically stable than the Pleistocene. The archaeological period that coincided with early the **Holocene** is the **Neolithic** (or “New Stone Age”), the period in which agriculture arose and human subsistence changed dramatically in many parts of the world.

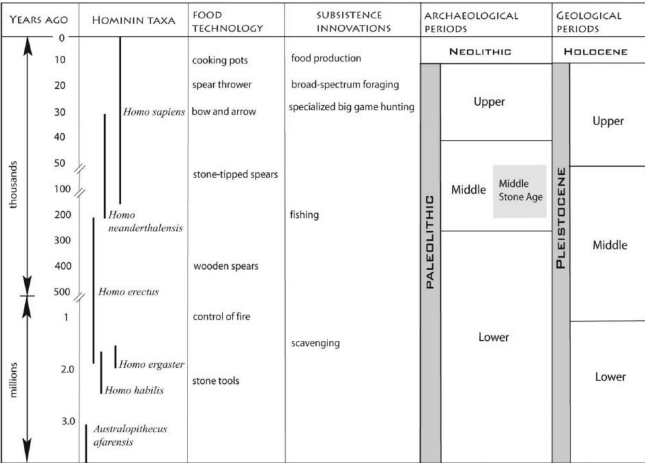


FIGURE 3.1. Comparative chart showing archaeological and geological periods and associated developments in food acquisition and food technology.

Figure 3.1 is from Gremillion, showing archaeological versus geological periods again time, hominin species present, and innovations in food technology and subsistence practices.

Source: Gremillion

The **Pleistocene** is the geological epoch that lasted from approximately 2.6 million years ago to 11,700 years ago. This period is defined climatically, as it is the epoch dominated by glacial conditions, cold periods in which glaciers expanded. For this reason, the Pleistocene is often popularly termed the “Ice Age.” The entire Pleistocene was not uniformly cold, however, as glacial periods were interspersed with interglacial periods—warmer periods in which glaciers shrank. These fluctuations occurred over regular cycles related to different amounts of solar radiation that hit the earth due to long-term asymmetries in Earth’s orbit and axial tilt, as well as changes in solar output. During the Last Glacial Maximum, around 26,500 to 20,000 years ago, glaciers covered the northern half of North America and Europe.

The **Paleolithic** is the archaeological era that lasted from approximately 3.3 million years ago to 12,000 years ago. This period is defined as that from the earliest evidence of human-made stone tools (the earliest technology discovered to date) until the advent of farming around 12,000 years ago. Note that this makes it roughly coincident with the Pleistocene, but the Paleolithic is defined by technology, not by climate (though the two are closely linked).

The **Holocene** is the geological epoch that lasted from approximately 11,700 years ago up to the near-present, the time at which humans began to affect the climate cycle so greatly as to create a new geological era, proposed to be named the Anthropocene.

The Holocene is defined climatically, as it differs markedly from the preceding Pleistocene, with its glacial periods. The Holocene has been warmer and wetter, with smaller glaciers, higher sea levels, and more stable climatic conditions than the Pleistocene exhibited.

The **Neolithic** is the archaeological era that began approximately 12,000 years ago; it ended at different times in different parts of the world. This period is defined as that in which advanced stone tools were the dominant form of durable technology, beginning with early agriculture and ending with the introduction of metal technologies. The earliest copper tools appeared in Southwest Asia around 6,500 years ago, but not until 4,000 to 3,000 years ago in other parts of Eurasia, and only after European contact in the Americas. Note that the entirety of the Neolithic took place within the Holocene; it has been argued that agriculture only became possible in the Holocene due to the warmer, wetter, and more stable climate of that epoch.